

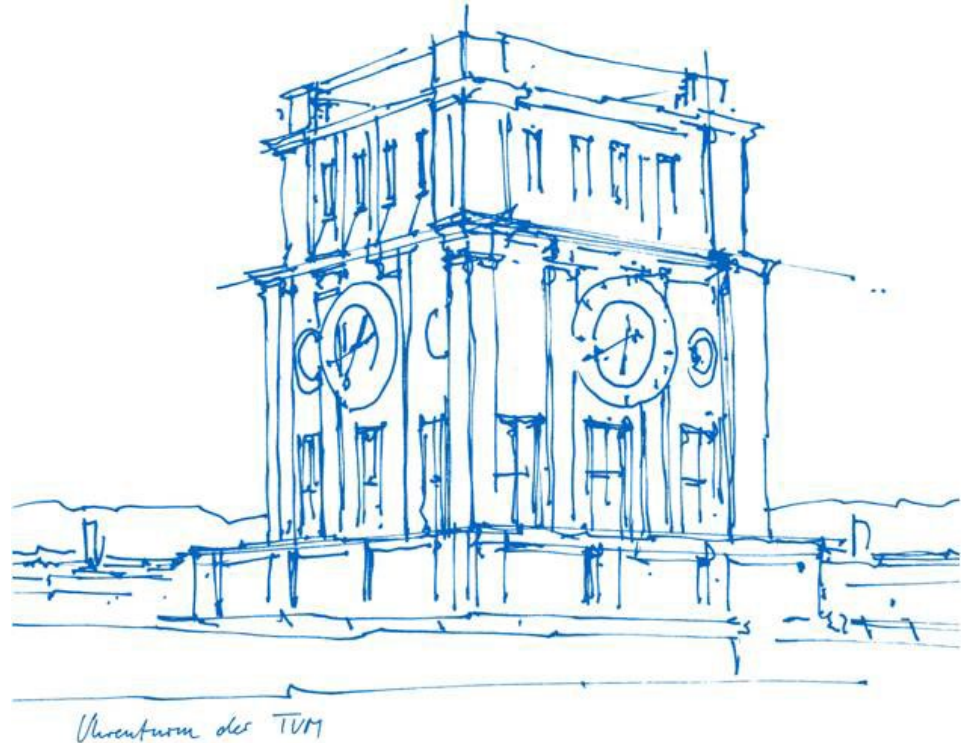
Geo-Information

Introduction to the Exercises

Winter Semester 2025 / 2026

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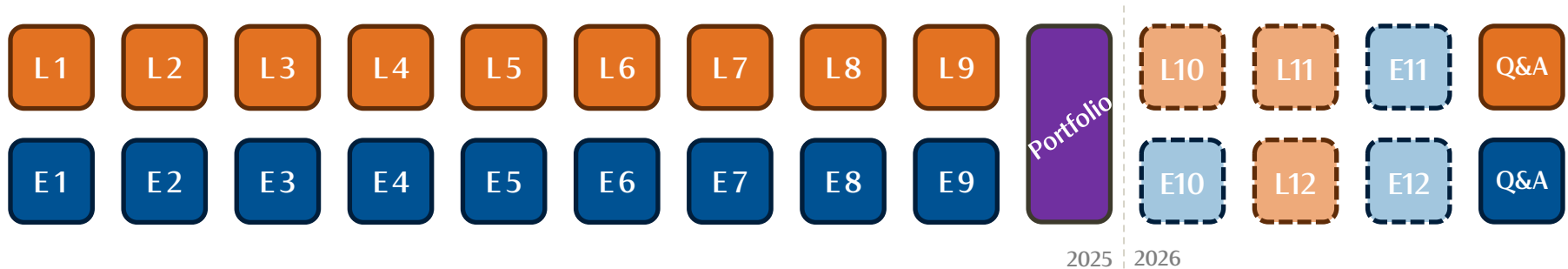
Welcome to the Module Geo-Information / Geoinformation

- BV300003 Geo-Information (M.Sc. Cartography, 6 ECTS)
- BGU30062 Geoinformation (M.Sc. ESPACE, 5 ECTS)

➤ What is 1 ECTS? How much is 1 ECTS?

- 1 ECTS credit equals 30 working hours -> 2 working hours per week (SWS)
 - 6 ECTS -> 12 SWS (4 SWS of Contact Hours + 8 SWS of Self-Study Hours)
 - 5 ECTS -> 10 SWS (3 SWS of Contact Hours + 7 SWS of Self-Study Hours)

Module Structure: Lectures with Integrated Exercises



- BV300003 Geo-Information (M.Sc. Cartography, 6 ECTS): All the planned classes.
- BGU30062 Geoinformation (M.Sc. ESPACE, 5 CETS): **Lecture/Exercise 10 – 12** are optional.

Overview of the Exercises



Basic Topics

E0: Getting Started

E1: Selecting and Editing Features

E2: Symbolization in a GIS

E3: Georeferencing and Modifying Features

E4: 3D Visualization and Animation

E5: Spatial Analysis Methods

E6: Vertical and Hydrological Analysis

E7: Network Analysis

E8: Spatial Data Integration

E9: Cartographic Generalization

Portfolio: Map Design in a GIS

Advanced Topics

E10: QGIS and Open GIS

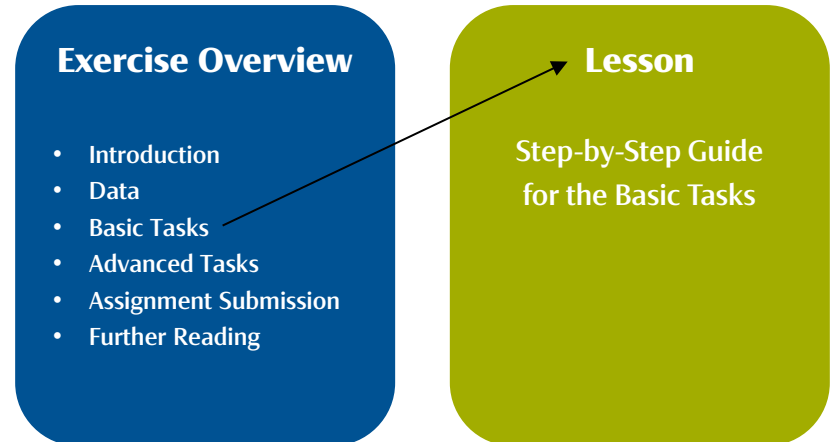
E11: Open Geospatial Data I

E12: Open Geospatial Data II

By Dr. Wangshu Wang

Agenda for Each Exercise

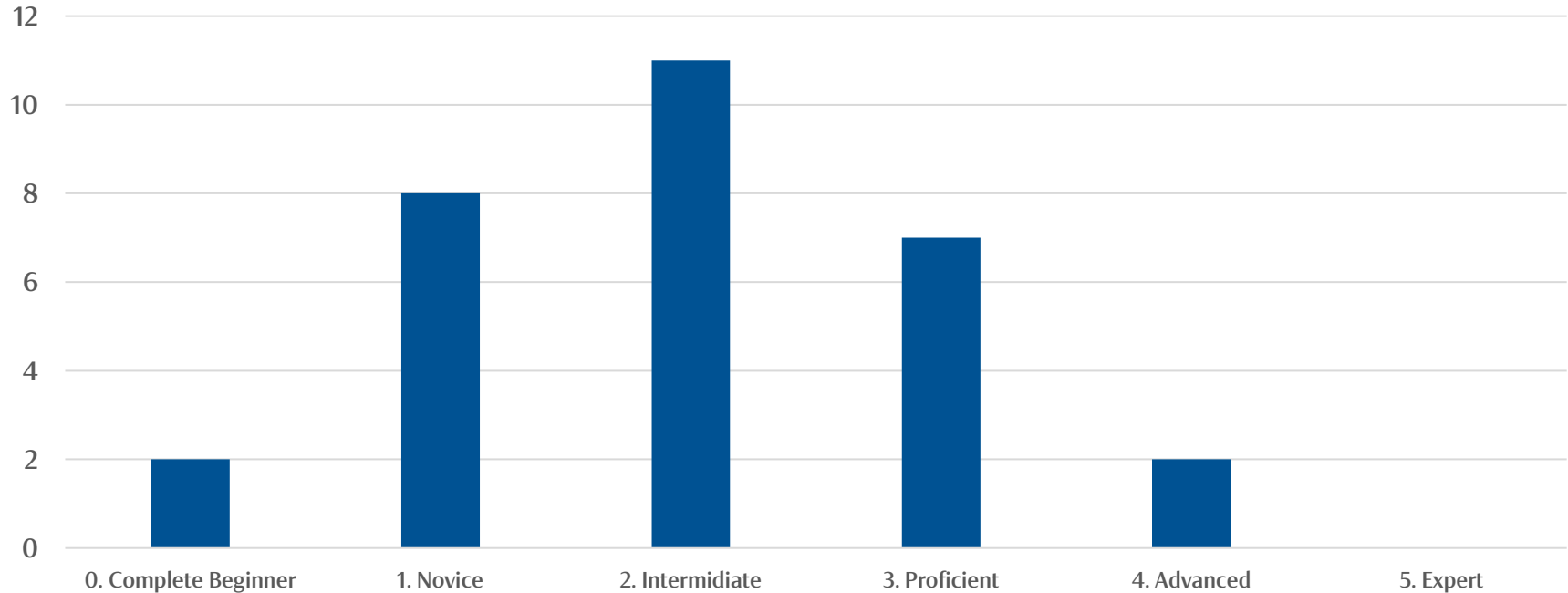
- **Wrap-Up:** Quizzes and Reviews related to Lecture and Exercise Content
- **Introduction:** to each Exercise
- **Exercise Time:** with the Materials in the Course Portal



Assignment Submission

- After some exercises (i.e., E2, E4, E8, E9, P, E10, E12), you need to make assignment submissions on **Moodle**.
- You will receive (if there is) feedback on your assignment submissions.
- The assignment submissions will not be graded.
- Students who **successfully complete all exercise assignments** will receive a **bonus** (+0.3 on the final grade if the exam is passed).

Pre-Course Survey: How is your current Desktop GIS Software skill?



User Manual on the Exercise Material

- Read through the **Exercise Overview** page, check your familiarity with each **Basic Task**:
 - **If you are new to this task:**
 - Follow the Lesson page instructions step by step.
 - The Advanced Tasks can be skipped.
 - **If you have an idea of the workflow but cannot reproduce it:**
 - Use the Lesson page instructions to update/refresh your knowledge.
 - Proceed to the Advanced Tasks.
 - **If you can reproduce the workflow without checking the Lesson page:**
 - Move on to the Advanced Tasks.
 - Achieve the same results using QGIS.

Basic Tasks

- ☐ **Creating a New Project** (1.2)
 - Connecting a Folder to the Project (1.2.1)
 - Adding a Database (1.2.2)
- ☐ **Adding a Layer**
 - Adding a Vector Layer (1.3)
 - Adding a Raster Layer (1.4)
- ☐ **Selecting Features**
 - Using a Pop-Up to Display a Feature's Attributes (1.5)
 - Selecting by Interactive Feature Selection (1.6)
 - Selecting by Location (1.7)
 - Selecting by Attributes (1.8)
- ☐ **Creating a New Feature Class** (1.10)
- ☐ **Editing New Features** (1.11)
 - Creating New Features (1.11.1)
 - Creating a New Template for Feature Creation (1.11.2)
- ☐ **Editing Fields and Attributes of Features** (1.9, 1.12)
 - Adding New Fields (1.12.1)
 - Renaming Fields (1.9.1)
 - Editing Fields (1.12.2)
 - Converting a String Field to a Numeric Field (1.9.2)
 - Calculating Fields (1.12.3)
 - Combining Editing Skills (1.12.4)

User Manual on the Exercise Material

The most common case would be:

If you have an idea of the workflow but cannot reproduce it.

- **Need to refresh your memory**
 - Use the Lesson page to recall specific steps or tools.
- **Are switching from ArcGIS Desktop (ArcMap)**
 - Focus on learning the new ribbon interface and updated workflows.
- **Are coming from QGIS or other GIS software**
 - Use the Lesson page to learn ArcGIS Pro's specific tools and terminology.
- **Are new to the English interface**
 - Following the steps will help you become familiar with the terms.
- **Are self-taught**
 - Use this as an opportunity to align your methods with a standard workflow.

Learning Outcomes of Exercises

- Implement the **concepts** of Geoinformation into practice.
i.e., spatiotemporal representation, spatial data analysis, cartographic techniques.
- Develop robust skills for **geodata processing and analysis**.
i.e., spatial analysis, network analysis, vertical and hydrological analysis, spatial data integration, etc.
- Acquire **professional cartographic skills** for map design and visualization.
i.e., map symbolization, map layout, 2D and 3D map generalization, map design principles, etc.
- Gain practical experience in using **ArcGIS Pro** for various GIS tasks and reach an **advanced skill level**.
- Become familiar with the capabilities of **open-source GIS software (QGIS)** and the processes for working with **Open Geodata (OpenStreetMap Data)**.

How should your Desktop GIS Software skills be after the course?



Pre-Course Survey: Your Expectation on this Module

- ☐ Start from the very basic +
- ☐ Practical with real-world use cases +
- ☐ Learn ArcGIS Pro with QGIS Experience +
- ☐ Learn something new/advanced from ArcGIS Pro +
- ☐ Update knowledge in English +
- ☐ Refresh/Dust off and update knowledge +
- ☐ Learn QGIS +
 - [E10]
- ☐ Open-source datasets +
 - [E10 – E12]
- ☐ All possible GIS tools -
 - We focus on Desktop GIS Software, especially ArcGIS Pro.
- ☐ Imagery / Raster Data -
- ☐ GIS with Programming -
- ☐ Geodatabase -
- ☐ GeoData Mining -
 - These topics will be mentioned, but they are not the focus.
 - Visit other courses to learn more.

Examination Method

- A **60-minute, closed-book written exam** at the end of the semester.
- Both **lectures and exercises** are covered in the exam. Quizzes will be helpful.
- Exercise assignment submissions are not graded.
- The **successful completion of all exercise assignments** will result in a **bonus**.
- The content of Lectures/Exercises 10–12 will only be covered in the exam for the Module BV300003 Geo-Information (M.Sc. Cartography, 6 ECTS).

Course Principles

- **It's YOUR course.** Be active and flexible to really learn something.
- If you have anything dissatisfied or/that can be done better, **just let me know asap.**
- If you find any errors or outdated content in the course portal, **please tell me! Much appreciated!**
- Please **be punctual** and respect others' time. Otherwise, you will miss the quiz session.
- We have 45+ students. **Help each other!**
- Please **keep your volume down** and respect others.
- Do not simply follow the 'Lessons' mechanically, **learn smart! ChatGPT can help!**

Any Questions?

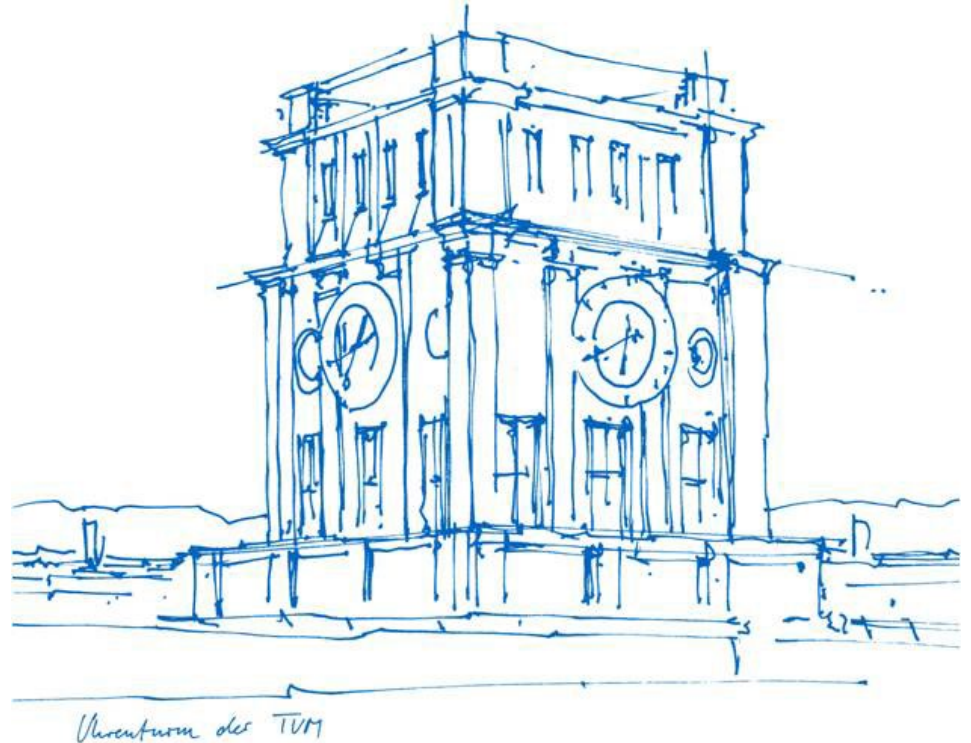
Geo-Information

E1 – Selecting and Editing Features

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Desktop GIS Software



Commercial



Open Source



Imagery-Focused (Desktop GIS) Software

- Agisoft Metashape
- ENVI
- ERDAS IMAGINE
- SNAP (Sentinel Application Platform)
- Google Earth Engine
-



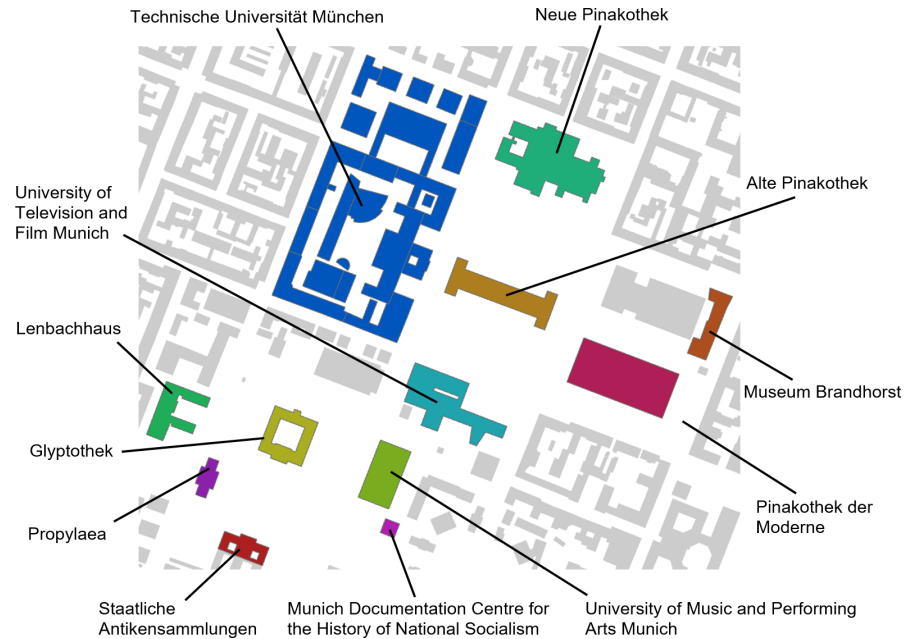
The Exercise Platform: ArcGIS Pro

- The most dominant commercial GIS software in the world by Esri.
- It has modern design, standardized workflow, stable performance, and detailed official documentation.
- TUM provides you with the campus license.
- It's a **Windows only software**. For the users of MacOS or Linux,
 - Use the public computers in computer labs (CIP-Pools), like 0712, N0199, 3209, 3238, etc.
 - Install a Windows virtual machine on your PC, via [Parallels Desktop](#) or other virtualization software.
 - Use a cloud-based Windows virtual machine, via [Microsoft Azure](#) or other cloud platforms.

Save Data Safely on Your Personal Webdisk

- If using a TUM PC, always save everything on your **personal webdisk**.
- Access your webdisk in a browser at <https://webdisk.ads.mwn.de/>
- Map Network Drive on your PC at [\\nas.ads.mwn.de\ge32kiw](https://nas.ads.mwn.de/ge32kiw) (Windows OS)
 - On MacOS, use **smb://nas.ads.mwn.de/ne23mek**
 - More information can be found [here](#) (Windows OS) or [here](#) (MacOS)

Study Area of E1 –E5



 **Kunstareal
München**

Basic Tasks of E0 & E1

- **E0 – Getting Started**
 - (Installing and) Starting ArcGIS Pro with your Campus License
 - (If using a TUM PC) Save Data safely on your Personal Webdisk
 - The Dataset and the Study Area
- **E1 – Selecting and Editing Features**
 - Creating a new Project and Adding Layers
 - Selecting Features in multiple ways
 - Creating a new Feature Class, Editing new Features and their Attributes

Exercise Time!